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A system dynamics model of port hinterland intermodal transport: A case study of Guangdong-Hong Kong-Macao Greater Bay Area under different carbon taxation policies

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ABSTRACT

The port cluster in the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) is an important network hub. It not only connects domestic and international markets, but also plays a vital role in stimulating the local and global economy. However, the GBA is facing the challenges in its port hinterland intermodal system due to its surging transport demands. The optimization and integration of the two goals of economy and eco-environmental protection must be considered at the same time. This study makes an endeavor to establish a port hinterland intermodal transport System Dynamics (SD) model of the GBA. The simulation results show that the carbon taxation policy is a relatively direct and effective incentive for inducing the modal split of port hinterland intermodal to a greener mode. Under the same carbon tax rate, the emission reduction effect in descending order is Shenzhen Port, Guangzhou Port, and Hong Kong Port, which means Shenzhen Port is the most sensitive to carbon tax rate due to its huge emission reduction potential. To achieve the national emission reduction goal, the carbon tax rate of BGA needs to be set at the medium to high level. Nevertheless, local governments still have to formulate carbon taxation policies based on local conditions.

1. Introduction

The Guangdong-Hong Kong-Macao Greater Bay Area (GBA) concept was proposed in 2015 and it was better known as Pearl River Delta (PRD). It is becoming the fourth active bay area for global economic activities and owning three world-class ports in the GBA, namely Shenzhen Port (4th), Guangzhou Port (5th) and Hog Kong Port (9th) according to Lloyd's List's seaport ranking 2022, in terms of container throughputs. As a hub for connecting domestic and international markets, this three-port-cluster plays an important role in the development of regional economy(Deng, Lu, & Xiao, 2013; Kong & Liu, 2021). These three ports have made positive contributions not only to the port cities but also to the surrounding regional economy. These ports and their hinterlands linked with intermodal transportation system, e.g., inland roadway, railway and waterway. The competitiveness of a seaport depends on the extent the cargo handled in the port can reach its hinterland destination or come from its hinterland origin (Bergqvist, Macharis, Meers, & Woxenius, 2015). The competitiveness of the port no longer depends only on its internal performance, but more on its hinterland transport ability and the fight for cargo sources inside hinterland (Li, Qu, & Ma, 2020; Sdoukopoulos & Boile, 2020; Zhang, Wang, Peng, & Guo, 2017). Therefore, a sustainable hinterland intermodal transportation system is not only a necessary aspect for ports to maintain competitiveness, but also a tool of competition (Bergqvist & Woxenius, 2011). In the GBA, there is a high degree of overlap in the hinterland of ports, and the fight for cargo sources is fierce. More than 70% of the container throughputs of Guangzhou Port and Shenzhen Port come from the Pearl River Delta (Long, Lixin, Zhongping, Hualian, & Zhichao, 2019), and more than 65% of the container traffic volume between Hong Kong and the mainland of China comes from the Pearl River Delta (Census And Statistics Department, 2019). Therefore, it is meaningless to study the port hinterland intermodal transport system from the perspective of a single port. The impact of the high overlap of port hinterland and competition from other ports on the hinterland intermodal transportation system should be taken from the perspective of the port cluster (Li, Haralambides, & Zeng, 2022). Free trade (external factor) and infrastructure development (internal factor) have led to heightened competition among neighboring, regional ports, and the GBA is no exception. This phenomenon leads to unnecessary duplication of effort and waste of scarce port resources (Chen et al., 2020; Ma, Jia, She,

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Haralambides, & Kuang, 2021). Chen et al. (2020) called for a port planning guidance for these major ports in the GBA, which could facilitate the coordinated, orderly and differentiated development of regional port industries. Hence, Hong Kong Port, Shenzhen Port and Guangzhou Port should have their differentiated future development strategies.

The International Energy Agency report shows that CO_2 emissions from transport sector accounts for 25% of global CO_2 emissions, second only to the electricity and heat sectors. The negative externalities caused by the transport sector are widely concerned around the world. So far, road transportation accounts for the largest share of negative externalities generated by transportation (Iannone, 2012). Hence, in response to the increasing emissions by road transportation, low-carbon footprint transport modals, including railway and waterway, have been encouraged for port hinterland intermodal transport by the government. For example, in the guidance issued by the Ministry of Transport of China on the construction of world-class ports, the proportion of railway and waterway hinterland transportation has been taken as an important evaluation indicator for the development of Chinese ports. Both the Outline Development Plan for the Guangdong-Hongkong-Macao Greater Bay Area (2019) and the Green Port Action Plan of Guangdong Province (2014–2020) are proposed to give priority to the development of railway and waterway transportation to reduce the CO_2 emissions of port hinterland intermodal transport and promoted the green development of the port cluster. As we know, many port authorities have developed ambitious strategies to foster their hinterland intermodal transportation (Bouchery, Woxenius, & Fransoo, 2020; Huang et al., 2021). On the one hand, hinterland connectivity can help increasingly shape port competitiveness. On the other hand, ports play important and socially responsible roles to mitigate carbon emissions via their hinterland intermodal network optimization(Guo, He, Lan, Yu, & Yan, 2022a, 2022b; Lam & Gu, 2013, 2016).

As the world's largest emitter, China's carbon emissions account for 28% of the global total, surpassing those of the US and Europe combined since 2012 (Z. Liu et al., 2021). In 2020, President XI further proposed that China should reach peak emissions before 2030 and achieve carbon neutralization before 2060. Hence, in the near future, China's economy might enter into a new phase of "carbon neutrality economy" since 2020. In China, transportation sector is an important source of carbon emissions, so reduction in carbon emissions in the transportation sector plays a key role in China's carbon peak and neutrality goals (Wang, Zhou, Zhu, Zhang, & Zhang, 2018). Over the past 30 years, China's port industry has developed by leaps and bounds and it plays an increasingly important role in its domestic economy and even in the global economy (Wang & Wang, 2019). Therefore, reduction of port carbon emission in the maritime realm is vital to achieve China's carbon neutral objective. To solve the negative externalities generated by transportation, financial penalties or incentives can be adopted, such as charges, taxes, and modal shift subsidies (Iannone, 2012). The carbon taxation as an economic policy instrument to solve negative externalities was first proposed by Arthur C. Pigou in 1920. It is a practical and effective measure to reduce carbon emissions, which has been adopted by Sweden, Finland, Denmark, Norway, the United Kingdom, Germany, Japan, France, and other countries. From the experience of these countries that have implemented carbon taxation policies, it can be seen that carbon taxation plays an important role in reducing CO_2 emissions (Carl & Fedor, 2016). Carbon taxation can reduce carbon dioxide emissions, but also have a negative impact on economic growth (Callan, Lyons, Scott, Tol, & Verde, 2009; Lu, Tong, & Liu, 2010). Therefore, not only the emission reduction effect of the carbon taxation policy but also the emission reduction cost caused by economic losses must also be taken into consideration. As the world's largest developing country and carbon emitter, China is facing the challenge to reduce CO_2 emissions and designing an appropriate carbon taxation policy for its transportation sector (Zhou, Fang, Li, & Liu, 2018). Before adopting the carbon taxation as a mitigation instrument, it will be beneficial to recognize the complexity of the port hinterland intermodal transport system and evaluate the potential impact of carbon taxation on the reducing CO_2 emissions and economic growth. Port hinterland intermodal transport is a dynamic system with multiple connection, information feedback, and causal interactions among multiple variables. Because of the complexity of the port hinterland intermodal transport system, it is worthwhile to apply system dynamics to predict the impact of carbon taxation policies in a simulated environment, so as to help policy-makers to plan ahead for changes in future policy directions.

The remainder of this paper is organized as follows. In Section 2, a concise literature review is provided. We introduce the System dynamics methodology and present the models in Section 3. The simulation results and sensitivity analyses are discussed in Section 4. Lastly, Section 5 is dedicated to conclusions and implications.

2. Literature review

A few decades ago, as early as in 1970s, some scholars proposed the similar notions, port hinterland-foreland continuum or port triptych (Charlier, 1992; Robinson, 1970). And these notions have been updated and extended in a relatively recent literature as a conceptual model named port regionalization (Notteboom & Rodrigue, 2005). The hinterland is defined as the continental area of origin and destination of traffic landward-side flows through a port (van Klink & van den Berg, 1998). In the new development phrase of port regionalization (Notteboom & Rodrigue, 2005), a port's hinterland can be divided into continuous hinterland and discontinuous hinterland, and hence an efficiently integrated intermodal system is very crucial in order to secure hinterland market share under an intense port competition environment (Caballe-Valls, de Langen, Garcia-Alonso, & Vallejo-Pinto, 2020). To connect the hinterland especially the discontinuous hinterland, ports make use of a variety of transport modals, such as road, railway, and waterway. A review paper summarized 50 research articles from 1972 to 2012 for the port hinterland intermodal container flow optimization models with green concerns and provided some suggestions about future research issues (Lam & Gu, 2013). Since then, there still exists a considerable amount of literature on the intermodal network optimization (Dai, Yang, & Li, 2018a, 2018b; Halim, Kwakkel, & Tavasszy, 2016), intermodal transport efficiency (Woodburn, 2017; Zhen, Wang, Wang, & Qu, 2018) and intermodal service chain integration and choice (Franc & Van der Horst, 2010; Talley & Ng, 2017), etc.

Moreover, significant attention on port hinterland intermodal transport research is given to inland roadway, waterway and railway transport separately (Braekers, Caris, & Janssens, 2013; Caramuta, Longo, Montrone, & Poloni, 2021; Konings, van der Horst, Hutson, & Kruse, 2010; Meers, Macharis, Vermeiren, & van Lier, 2017; Munim & Haralambides, 2018; Wang, Meng, & Miao, 2016; Woxenius, 2007; Woxenius & Bergqvist, 2011), whilst few studies talk about the integration of these three modals together (Dai et al., 2018a, 2018b; Shi & Li, 2016; Caris, Limbourg, Macharis, van Lier, & Cools, 2014; Wiegmans & Konings, 2015). In common practices, the purpose of port hinterland intermodal network optimization is promoting modal shift from road to rail or barge transport. However, some research papers have proved that blindly maximizing the modal shift is harmful for both cost and carbon emissions, and there exists a carbon optimal level of modal shift for different scenarios (Guo et al., 2022a, 2022b; Bouchery & Fransoo, 2015). Taking the Shanghai port and the Yangtze River Delta (YRD) hinterland as a case , the recommended target ratios can be set a balance of 5:3:2 for road, water, and rail (Guo et al., 2022a, 2022b). When carbon emission issues (external costs) are concerned in these papers, a series of policy packages can show better performance compared with a single policy type (Wu & Zhang, 2022). Carbon tax, as a commonly implemented policy measure, not only as an effect on environmentally friendly modal shift, but can severely impact the competitive situation of port hinterland intermodal infrastructure components (Gleser, Elbert, & Wu, 2023; Huang, Han, & Jiang, 2021). Government money spent on

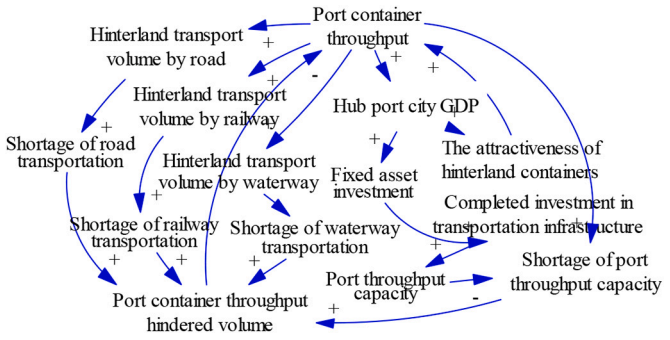


Fig. 1. The causal loop diagram of system dynamics model for port hinterland intermodal transport system.

intermodal infrastructure and operational subsidies must be aligned with an understanding of how intermodal flows are embedded within internal and external relationships, and hence the logistics platform is very important as well (Monios, 2015). After a careful literature review, we can find that deterministic and static operations research models are commonly applicable to investigate the port hinterland intermodal transport systems and related issues. However, these port hinterland intermodal transport systems are often complex and dynamics, typically with a nonlinear fashion (Macharis & Bontekoning, 2004). Some simulation research methods or tools can be served as indispensable supplements, such as the system dynamics method.

System dynamics (SD) was first proposed to analyze complex dynamic feedback systems by Forrester from MIT in the 1950s and 1960s, which can analyze the relationship between various factors and simulate quantitative data (Forrester, 1958). This makes it easier for us to understand the system as a whole and formulate various relevant policy scenarios to control the dynamic evolutionary mechanism of the system (Yuan, Ji, Chen, Chen, & Chen, 2008). From the perspective of system dynamics, the port hinterland intermodal transport system, as a component of the port hinterland economic region system, is affected by the society, economy, and environment (Hou & Geerlings, 2016a, 2016b). The coordinated development of the port hinterland intermodal transport system is a dynamic evolutionary process. Therefore, studying port hinterland transport modals and CO_2 emissions through scientific and systematic methods can provide a basis for the design of carbon taxation policy. It is obviously difficult to solve such problems using traditional linear system theoretical methods, while system dynamics is very effective in analyzing the structure and behavior of complex systems. Back in 1994, Abbas and Bell first confirmed that SD can be used for transportation modeling, and they listed 12 advantages of this approach compared to traditional transport modeling approaches (Abbas & Bell, 1994). Currently, SD has been widely used in various research fields, including social and economic system research (Grygorkiv & Grygorkiv, 2015), ecosystem research (Tan, Yang, Château, Lee, & Chang, 2018), and transportation research (Suryani, Chou, & Chen, 2010) and so on. It is worth emphasizing that system dynamics in port research mostly focuses on port-city system (Li, Zhang, Lin, & Huang, 2019), port service (Fu, Cao, Liang, & Wu, 2018), risk management (Kwesi-Buor, Menachof, & Talas, 2019; Yeo, Pak, & Yang, 2013), port economy (Xiu & Zhao, 2021) and port congestion (Muravev, Rakhmangulov, Hu, & Zhou, 2019; Xu, Li, Liu, & Yang, 2021), and rarely involves port hinterland intermodal transport. Besides, although system dynamics has largely been used in transport research literature to analyze future policy instruments or measures to reduce CO_2 emissions, it has not been used much in measuring the effectiveness of carbon taxation on the port hinterland intermodal transport system. For instance, Cheng, Chang, and Lu (2015) and Liu, Ma, Tian, Jia, and Li (2015) developed the urban passenger transport SD model to simulate the effects of urban transportation management policies on CO_2 emissions. Gupta, Bandyopadhyay, and Singh (2019) developed the road passenger transport SD model to examine whether carbon taxation policy could be effective in reducing CO_2 emissions. We can also discover that even though there exist some recent research articles to investigate the port hinterland intermodal system using system dynamics approach, but their case study areas do not include the GBA (Pearl River Delta) area, only including the Yangtze River Delta and Bohai Bay areas (P. Liu et al., 2021; Liu, Liu, Du, & Mu, 2019; Hou & Geerlings, 2016a, 2016b). Hou and Geerlings (2016a, 2016b) found that compared with other measures (such as technological and spatial measures), the modal shift measures (shifting freight from road to railway and waterway) show greater potential to reduce CO_2 emissions from port hinterland intermodal transport. Such similar conclusions have been verified in other system dynamics papers (P. Liu et al., 2021; Liu et al., 2019).

In addition, it can be found that all these SD models focused on researching the effects of policies on reducing CO_2 emissions, without considering the cost of emission reduction.

Therefore, to fill the potential research gap of the impact of carbon taxation on the hinterland intermodal transportation, this study makes an endeavor to establish a port hinterland intermodal transport SD model of the GBA and simulates the characteristics and trends of the port hinterland transport modals and CO_2 emissions in different port area. The developed SD model attempts to assess the effectiveness of the carbon taxation policy to mitigate CO_2 emissions from hinterland intermodal transport, taking into account emission reduction costs.

3. Methodology

The SD model consists of two parts. The first part is a causal–loop diagram, which describes a set of simplified causal relationships between different systems developed during the model construction. The second part is a flow diagram representing the quantitative relationship among variables. In addition, error analysis based on historical statistical data should be evaluated to ensure the accuracy and validity of the prediction results. This proposed SD model in our research is built using the Vensim PLE® software package (Ventana Systems, Inc., Harvard, MA, USA).

3.1. Model hypotheses

This study aims to establish a sustainable development SD model to assess the impact of carbon taxation policy on the emissions of the GBA hinterland intermodal transportation. Based on the research objective, the following hypotheses are adopted.

(1) According to the 2019 Macau Yearbook of Statistics, the transportation industry only accounts for 3.9% of Macau's industry. In 2019, Macau's container throughput completed 133,038 TEU, accounting for only 0.16% of the container throughput of the GBA. Therefore, Macau is not included in the SD model.
(2) The model in this article focuses on the carbon emissions from transportation between the port and the hinterland, regardless of the carbon emissions from transshipment and other sectors. This is because CO_2 emissions of port hinterland intermodal transport mainly come from direct transportation, while the emissions from transshipment and other sectors are very small and difficult to measure (Liu et al., 2019).

3.2. Causal loop diagram

The causal loop diagram (CLD) was constructed to clarify the relationships among the main variables in the system (Fig. 1). These loops in CLD have two directions. The positive feedback loops have a strengthening effect, while the negative feedback loops have a balancing effect. Positive links are denoted with a "+", and negative links are denoted with a "-". In Fig. 1, a virtual variable name is not the real

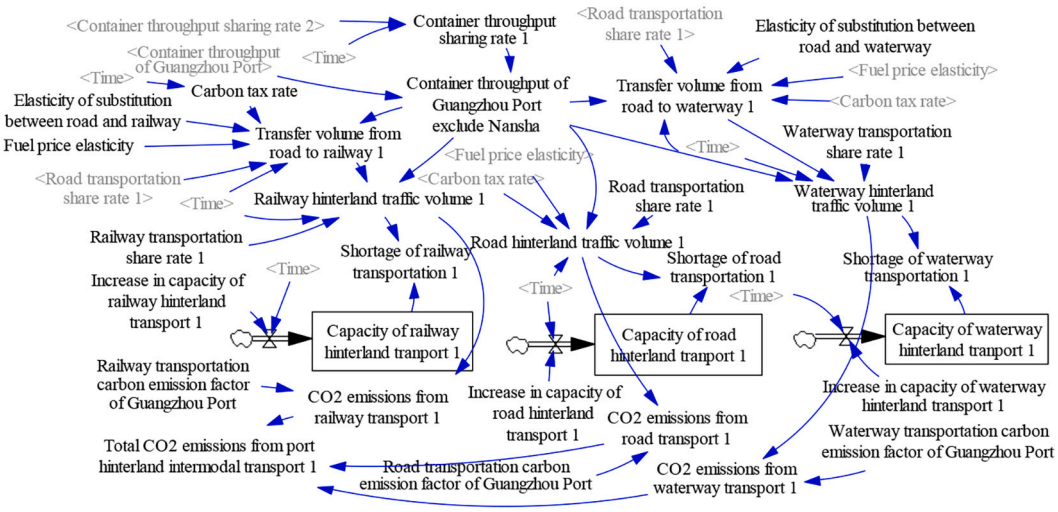


Fig. 2. The flow diagram of Guangzhou Port (exclude Nansha) intermodal transport subsystem.

variable name, and it only represents the name of a variable of the same class and attribute. This causal loop diagram with virtual variable names can help us to understand the causal logic of different variable classes.

3.3. Flow diagram

Based on the results of the causal loop diagram analysis, the factors in each loop are described in more detail by introducing state variables, rate variables and auxiliary variables. The flow diagram model is constructed, as shown in Appendix A ~ C (A is an integrated diagram while B is from three ports' perspectives and C is from seven cities' perspectives). According to the statistics of the 2020 China Port Yearbook, about 60% of the container throughput in the Nansha Port Area of Guangzhou are shipped via inland waterway, while more than 60% of the container throughput in the Huangpu Port Area and the Xinsha Port Area are transported via inland roadway, and there are obvious differences in their intermodal structures of different port areas of even the same port. The same situation exists in Shenzhen Port, more than 80% container throughput are shipped via road in its Eastern Port Area, while this ratio is less than 60% in its Western Port Area and more than 40% container throughput is still transported using inland waterway in its Western Port Area. Therefore, when setting up the subsystems, it is necessary to distinguish the characteristics of the intermodal structures of different port areas, so as to present the development status of the port cluster intermodal system in the GBA to the greatest extent. In this study, the SD model includes nine subsystems: hinterland economy subsystem, Guangzhou Port (exclude Nansha) hinterland intermodal transport subsystem, Guangzhou Port (Nansha) hinterland intermodal transport subsystem, Shenzhen Port (Eastern) hinterland intermodal transport subsystem, Shenzhen Port (Western) hinterland intermodal transport subsystem, Hong Kong Port hinterland intermodal transport subsystem, Guangzhou urban economy subsystem, Shenzhen urban economy subsystem and Hong Kong urban economy subsystem.

(1) Hinterland economy subsystem

The typical characteristic of the port cluster in the GBA is that the hinterland has a high degree of overlap. There is a complex relationship of competition and cooperation between the ports. Therefore, it is necessary to adopt a reasonable method to allocate hinterland containers among the ports. With reference to the basic trade gravity model established by Tingbergen (1962), the bilateral trade volume between two countries or regions is directly proportional to the total economic volume of the two countries and inversely proportional to the distance between the two countries, as shown in formula (1).

$$P_{ijt} = \frac{T_{jt}}{D_{ij}^2} \tag{1}$$

P_{ijt}: Hinterland attractiveness index; T_{jt}: GDP of port j in period t; D_{ij}: Distance between hinterland i and port j. Then, calculate the ratio of the attraction X_{ijt} of port j to hinterland i among all ports, as shown in formula (2) and Appendix A ~ C (A is an integrated diagram while B is from three ports' perspectives and C is from seven cities' perspectives). According to the attractiveness ratio X_{ijt}, the hinterland container volume was allocated to Guangzhou Port, Shenzhen Port and Hong Kong Port (Appendix A ~ C).

$$X_{ijt} = \frac{P_{ijt}}{\sum_j P_{ijt}} \tag{2}$$

As we know, the basic trade gravity model has been evolved and updated in many versions in the past few years. In short, the geographical distribution of a given country's trade flows with partner countries is determined by three sets of factors: (1) export supply, e.g. captured by GDP of the exporting country; (2) import demand, as given by GDP of the importing country; (3) and other factors that influence trade either negatively, such as transport costs, (typically proxied by distance), or positively, e.g., bilateral policy agreements that reduce cross-border barriers to trade (Stack, 2009).

In our research models, Guangzhou Port, Shenzhen Port and Hong Kong Port belong to the same GBA area, and the corresponding regional economies have the similar GDP volumes, for example, in 2019 (before COVID-19), Shenzhen's GDP is 2699 billion RMB, Guangzhou's GDP is 2363 billion RMB, while Hong Kong's GDP is 2492 billion RMB. We make an assumption that Shenzhen, Guangzhou and Hong Kong have the same context settings in our research, although we know that there exists a trade border between Hong Kong and mainland China. Therefore, we apply the basic trade gravity model established by Tingbergen in 1962.

(2) port hinterland intermodal transport subsystem

The port hinterland intermodal transport subsystem includes five parts: Guangzhou Port (exclude Nansha), Guangzhou Port (Nansha), Shenzhen Port (Eastern), Shenzhen Port (Western) and Hong Kong Port hinterland intermodal transport subsystems. Variables in Guangzhou Port (exclude Nansha) intermodal transport subsystem can be divided into three categories: container throughput, transport modals and carbon emissions (Fig. 2). The increase in port container throughput will

Table 1
Each Port's Intermodal Carbon Emission Coefficient.

Port Carbon Emission Coefficient	Guangzhou Port	Shenzhen Port	Hongkong Port
Inland Roadway Carbon Emission Coefficient	81.14	67.18	94.23
Inland Railway Carbon Emission Coefficient	9.27	6.10	–
Inland Waterway Carbon Emission Coefficient	3.99	3.16	2.73

Source: compiled by the author based on the calculated results.

stimulate the demand for port hinterland intermodal transport, leading to increased pressure on road/railway/waterway transportation. The shortage of port hinterland intermodal transport depends on the difference between transportation demand and transportation capacity, which will inhibit the growth of port container throughput. Assuming that the current level of fuel efficiency will continue to persist for some time, the introduction of the carbon taxation policy will increase fuel price and lead to a reduction in consumption of fuel. The current model takes −0.8 long-term price elasticity of fuel suggested by Drollas (1984), Graham and Glaister (2002). In addition, Andersson (2019) proved that the carbon tax elasticity of demand for fuel is 3.1–4.5 times larger than the price elasticity. Therefore, a 10% increase in fuel prices will lead to a 36% reduction in fuel consumption at most and inhibition high fuel consumption transportation activities. The method of estimating port hinterland intermodal transport emissions is to multiply the carbon emission factor per unit by the amount of transported containers. Carbon emissions are commonly measured using the 2006 IPCC Guidelines for National Greenhouse Gas Inventories issued by the United Nations Intergovernmental Panel on Climate Change (IPCC), and there are "top-down" and "bottom-up" two approaches. Among them, the "top-down" approach is based on the fuel consumption of various means of transportation, and the total carbon emissions obtained by converting the carbon emission coefficients of different fuels. In our research, we apply the "top-down" approach. The calculated factors such as unit carbon emissions (Cheng, 2020), average transportation distance and average container weight of each intermodal mode are input into the formula, and the carbon emission coefficient of each port each intermodal mode can be obtained and shown in Table 1.

The Guangzhou Port (Nansha) hinterland intermodal transport subsystem, Shenzhen Port (Eastern) hinterland intermodal transport subsystem, Shenzhen Port (Western) hinterland intermodal transport subsystem and Hong Kong Port hinterland intermodal transport subsystem are similar to the Guangzhou Port (Nansha) hinterland intermodal transport subsystem (Appendix D). However, according to the Hong Kong Statistical Yearbook, Hong Kong ceased operations of freight trains in 2010. Therefore, container transportation between Hong Kong and the mainland of China is only by roadway and waterway.

(3) Urban economy subsystem

Urban economy subsystem includes three parts: Guangzhou, Shenzhen, and Hong Kong urban economy subsystem. Variables in Guangzhou urban economy subsystem can be divided into three categories: economy and investment scale, container throughput and port throughput capacity (Fig. 3). Since the subsystem only considers port-related transportation activities and their impact, GDP growth depends on the GDP growth rate and the value created by port activities. The scale of urban GDP will affect the amount of infrastructure investment in the

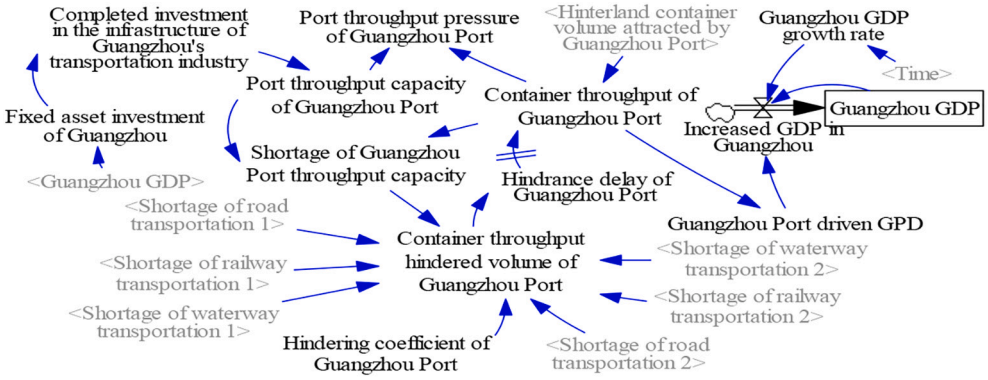


Fig. 3. The flow diagram of Guangzhou urban economy subsystem.

Table 2
The Validation of Simulation Results (Examined Variable: Container Throughput).

Year	Guangzhou Port (10,000 TEUs)			Shenzhen Port (10,000 TEUs)			Hong Kong Port (10,000 TEUs)		
	Actual value	Simulated result	Error	Actual value	Simulated result	Error	Actual value	Simulated result	Error
2010	1270.26	1296.39	2.06%	2251.00	2205.21	−2.03%	2369.00	2342.52	−1.12%
2011	1442.11	1424.22	−1.24%	2257.00	2307.08	2.22%	2440.00	2359.93	−3.28%
2012	1474.36	1462.19	−0.83%	2294.00	2364.77	3.09%	2309.00	2374.59	2.84%
2013	1550.45	1615.75	4.21%	2328.00	2280.67	−2.03%	2228.80	2158.30	−3.16%
2014	1662.62	1625.07	−2.26%	2404.00	2390.23	−0.57%	2228.30	2210.15	−0.81%
2015	1762.49	1750.58	−0.68%	2420.00	2426.65	0.27%	2011.40	1925.77	−4.26%
2016	1885.77	1830.98	−2.91%	2398.00	2458.67	2.53%	1981.30	2068.27	4.39%
2017	2037.20	2072.25	1.72%	2521.00	2557.63	1.45%	2076.30	2044.86	−1.51%
2018	2192.21	2280.01	4.01%	2574.00	2546.69	−1.06%	1959.60	1965.32	0.29%
2019	2323.62	2225.12	−4.24%	2577.00	2581.16	0.16%	1830.30	1875.80	2.49%
2020	2350.53	2453.99	4.40%	2655.00	2613.06	−1.58%	1796.00	1751.65	−2.47%
2021	2446.65	2568.48	4.98%	2877.00	2749.02	−4.45%	1780.00	1763.27	−0.94%

Source: compiled by the author based on the simulation results.

Table 3
The Validation of Simulation Results (Examined Variable: GDP).

Year	Guangzhou GDP (hundred million RMB)			Shenzhen GDP (hundred million RMB)			Hong Kong GDP (hundred million RMB)		
	Actual value	Simulated result	Error	Actual value	Simulated result	Error	Actual value	Simulated result	Error
2010	10,859.3	10,859.3	0.00%	10,002.2	10,002.2	0.00%	15,446.3	15,446.3	0.00%
2011	12,562.1	12,665.8	0.83%	11,807.2	11,884.4	0.65%	16,041.8	16,065.3	0.15%
2012	13,697.9	13,924.9	1.66%	13,319.7	13,487.5	1.26%	16,595.8	16,643.7	0.29%
2013	15,663.5	16,040.1	2.40%	14,979.5	15,250.9	1.81%	16,705.5	16,777.4	0.43%
2014	16,896.6	17,432.1	3.17%	16,449.5	16,827.4	2.3%	18,110.5	18,210.0	0.55%
2015	18,313.8	19,024.2	3.88%	18,014.1	18,511.6	2.76%	20,391.8	20,526.0	0.66%
2016	19,782.2	20,689.6	4.59%	20,079.7	20,719.2	3.18%	22,411.7	22,578.5	0.74%
2017	21,503.2	22,136.0	2.94%	22,490.1	23,292.4	3.57%	22,210.7	22,396.6	0.84%
2018	22,859.4	24,001.4	5.00%	24,222.0	25,175.6	3.94%	24,968.2	25,197.7	0.92%
2019	23,628.6	24,807.2	4.99%	26,992.3	28,076.4	4.02%	24,919.6	25,168.3	1.00%
2020	25,068.8	26,275.5	4.81%	27,759.0	28,114.4	1.28%	24,103.0	24,019.8	−0.35%
2021	28,232.0	29,011.4	2.76%	30,664.9	29,886.6	−2.54%	23,740.0	24,924.1	4.99%

Source: compiled by the author based on the simulation results.

Table 4
The Growth Rate of Port Container Throughput (No Carbon Tax Policy).

Port	13th Five-Year Plan (2016–2020)	14th Five-Year Plan (2021–2025)
Guangzhou Port	40.80%	27.40%
Shenzhen Port	8.39%	4.55%
Hongkong Port	−11.99%	−11.61%

Source: compiled by the author based on the simulation results.

Table 5
The Growth Rate of Port Container Throughput in the Different Scenarios. (The 14th Five-Year Plan Period, 2021–2025).

Port	LCT	MCT	HCT
Guangzhou Port	27.35%	27.01%	25.12%
Shenzhen Port	4.39%	3.39%	−1.33%
Hongkong Port	−11.65%	−11.93%	−12.89%

Source: compiled by the author based on the simulation results.

transportation industry, which in turn affects the port container throughput capacity. The shortage of port throughput and hinterland intermodal transport capacity will limit the growth of port container throughput, which in turn will affect the growth of GDP.

Shenzhen urban economy subsystem and Hongkong urban economy subsystem (Appendix E) are similar to Guangzhou urban economy subsystem. Moreover, the port throughput capacity of Hong Kong has not changed significantly in the past ten years. Therefore, in the Hong Kong urban economy subsystem, the port throughput capacity is only calculated by a table function, without considering the impact of fixed asset investment.

3.4. Authenticity test

To verify the authenticity of the proposed SD model, this study set the years between 2010 and 2025 as the simulation time, set "one year" as the simulation time step and observed the errors between the estimated values and its historical actual values. The examined variables included the container throughput of Guangzhou Port, Shenzhen Port and Hong Kong Port (Table 2). Another set of examined variables included Guangzhou GDP, Shenzhen GDP and Hong Kong GDP in Table 3. In Table 2 and Table 3, we only compare the actual values and simulated results from 2010 to 2021 to verify the effectiveness and robustness of this model. The simulation data from 2022 to 2025 are the forecasting values.

The errors between the values have been observed as lying under 5% which indicates that the model is reasonable and has the potential to be used for future policy analysis (Han & Hayashi, 2008).

4. Simulation results and discussion

Carbon tax policy is one of the effective policy tools to solve the negative externalities of carbon emissions through market mechanism means through price control, following the polluter-pays principle. The implementation of a carbon tax will have a wide and far-reaching impact on the social economy, energy consumption, ecological environment and many other aspects. Therefore, based on the system dynamics model constructed above, this section sets up multiple carbon tax simulation scenarios to investigate the impact of different tax rates on port hinterland intermodal systems and corresponding carbon emissions, and to provide decision-making references for promoting the sustainable development of port hinterland intermodal system and the implementation of carbon tax policies in the Guangdong-Hong Kong-Macao Greater Bay Area.

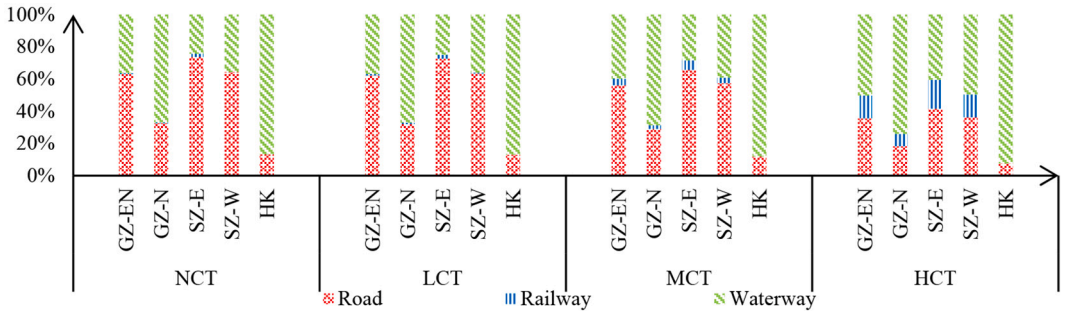


Fig. 4. The changes in share rate of different port hinterland transport modals.(Source: compiled by the author based on the simulation results.)

Table 6
Total CO_2 emissions and reduction effect in the different scenarios.

Scenario	Port Area	Total CO_2 Emissions from 2021 to 2025 (10,000 tons)	CO_2 Emission Reduction (10,000 tons)	Percentage Reduction in CO_2 Emissions
LCT	Guangzhou Port (exclude Nansha)	167.6204	2.4032	1.41%
	Guangzhou Port (Nansha)	312.5750	4.1869	1.32%
	Shenzhen Port (Eastern)	330.5828	5.2405	1.56%
	Shenzhen Port (Western)	291.0539	4.5821	1.55%
	Hong Kong Port	44.7151	0.5908	1.30%
MCT	Guangzhou Port (exclude Nansha)	153.0057	17.0179	10.01%
	Guangzhou Port (Nansha)	287.1139	29.6480	9.36%
	Shenzhen Port (Eastern)	298.9741	36.8492	10.97%
	Shenzhen Port (Western)	263.4188	32.2172	10.90%
	Hong Kong Port	41.1245	4.1814	9.23%
HCT	Guangzhou Port (exclude Nansha)	101.9715	68.0521	40.03%
	Guangzhou Port (Nansha)	198.1123	118.6496	37.46%
	Shenzhen Port (Eastern)	190.4604	145.3629	43.29%
	Shenzhen Port (Western)	168.5361	127.0999	42.99%
	Hong Kong Port	28.7591	16.5468	36.52%

Source: compiled by the author based on the simulation results.

4.1. Carbon taxation policy scenarios simulation

Based on the result of the authenticity test and the level of the carbon tax rate, this study established four basic scenarios, namely, no carbon tax (NCT) scenario, low carbon tax (LCT) scenario, medium carbon tax (MCT) scenario and high carbon tax (HCT) scenario, assuming that the carbon tax policy is to be implemented from 2022 (Liu, Lei, Xu, & Yuan, 2018; Liu, Wang, & Meng, 2023; WorldBank, 2020). Different scenarios are created by varying the tax rates to examine the impact on port container throughput, hinterland intermodal transportation and carbon dioxide emissions.

(1) NCT scenario

The NCT scenario is set in accordance with the current development situation, without carbon taxation policy. Settings related to nine subsystems are the same as above.

(2) LCT scenario

Chinese Academy of Fiscal Sciences once suggested that the carbon tax should start at a low level, with the initial rate of 10 RMB/tCO_2. Therefore, carbon tax rate is set to 10 RMB/tCO_2 in the LCT scenario.

(3) MCT scenario

According to the statistics of "State and Trends of Carbon Pricing 2020", the Carbon prices range from less than 1 \$/tCO_2 to 119 \$/tCO_2, with almost half of the covered emissions priced at less than 10 \$/tCO_2. Therefore, carbon tax rate is set to 70 RMB/tCO_2 in the MCT scenario.

(4) HCT scenario

The High-Level Commission on Carbon Prices estimated that carbon prices of at least 40–80 \$/tCO_2 by 2020 are required to cost-effectively reduce emissions in line with the temperature goals of the Paris Agreement. Therefore, carbon tax rate is set to 280 RMB/tCO_2 in the HCT scenario.

4.1.1. Impact of carbon taxation on port container throughput

According to the simulation data, the growth rate of container throughput in Guangzhou Port, Shenzhen Port and Hong Kong Port during the 14th Five-Year Plan period will be lower than that during the 13th Five-Year Plan period. Furthermore, the container throughput of Guangzhou Port will continue to maintain a high growth rate during the 14th Five-Year Plan period, reaching 27.4%. The container throughput of Hong Kong Port will continue to maintain a negative growth trend in the next five years (Table 4). In the LCT, MCT and HCT scenario, the changes in container throughput of Guangzhou Port and Hong Kong Port are less than 2%, while the maximum change in Shenzhen Port is nearly 5%. In contrast, Shenzhen Port is the most sensitive to carbon tax policies. Under the high-carbon tax scenario, the container throughput of Shenzhen Port will change from positive to negative growth during the 14th Five-Year Plan period (2021–2025), which will seriously affect the development of the port (Table 5).

4.1.2. Impact of carbon taxation on port hinterland intermodal transport modal and volume

The changes in the share rate of different hinterland intermodal transport modals are obtained using the SD model mentioned above, as shown in Fig. 4. In addition, GZ-EN, GZ-NS, SZ-E, SZ-W, HK represent Guangzhou Port (exclude Nansha), Guangzhou Port (Nansha), Shenzhen Port (Eastern), Shenzhen Port (Western) and Hong Kong Port respectively.

In the NCT scenario, the railway transport share rate of Guangzhou Port (exclude Nansha), Guangzhou Port (Nansha), Shenzhen Port (Eastern), Shenzhen Port (Western) and Hong Kong Port are 0.47%, 0.62%, 1.94%, 0.03% and 0.00% respectively. Although there have ports like Yantian, whose port hinterland transportation volume by railway ranks second in China, but its share rate of railway is still less than 2%. The waterway transportation share rate of Guangzhou Port (Nansha) is higher than that of Guangzhou Port (exclude Nansha). Similarly, the waterway transportation share rate of Shenzhen Port (Western) is higher than that of Shenzhen Port (Eastern). This is because Guangzhou Port (Nansha) and Shenzhen Port (Western) both have supporting barge transportation services-Guangzhou Port Shuttle Barge and South China Shuttle Barge, which make water transportation more efficient and convenient. To sum up, the port hinterland intermodal transport in the GBA is mainly by road and waterway, and rarely by railway. The main port hinterland transport modal of Guangzhou Port (exclude Nansha), Shenzhen Port (Eastern) and Shenzhen Port (Western) is by road. And the main port hinterland transport modal of Guangzhou Port (Nansha) and Hong Kong Port is by waterway. Moreover, even in the same sea port, the share rate of port hinterland transport modals in different port areas is quite different.

In the LCT, MCT and HCT scenario, it can be found that the carbon taxation policy has a depressing effect on road transportation. The low carbon tax is not effective in reducing road transportation volume, but the medium and high carbon tax will lead to a considerable reduction in road transportation volume. Comparing the five port areas in the model, the greater the share of road transportation in the port hinterland, the more obvious the inhibitory effect of the carbon taxation policy. Besides, the carbon taxation policy will induce the port hinterland transportation modal to change from road transportation to railway and waterway transportation. Compared with LCT scenario, MCT scenario and HCT scenario have a more significant effect on promoting the transformation of port hinterland transport modals.

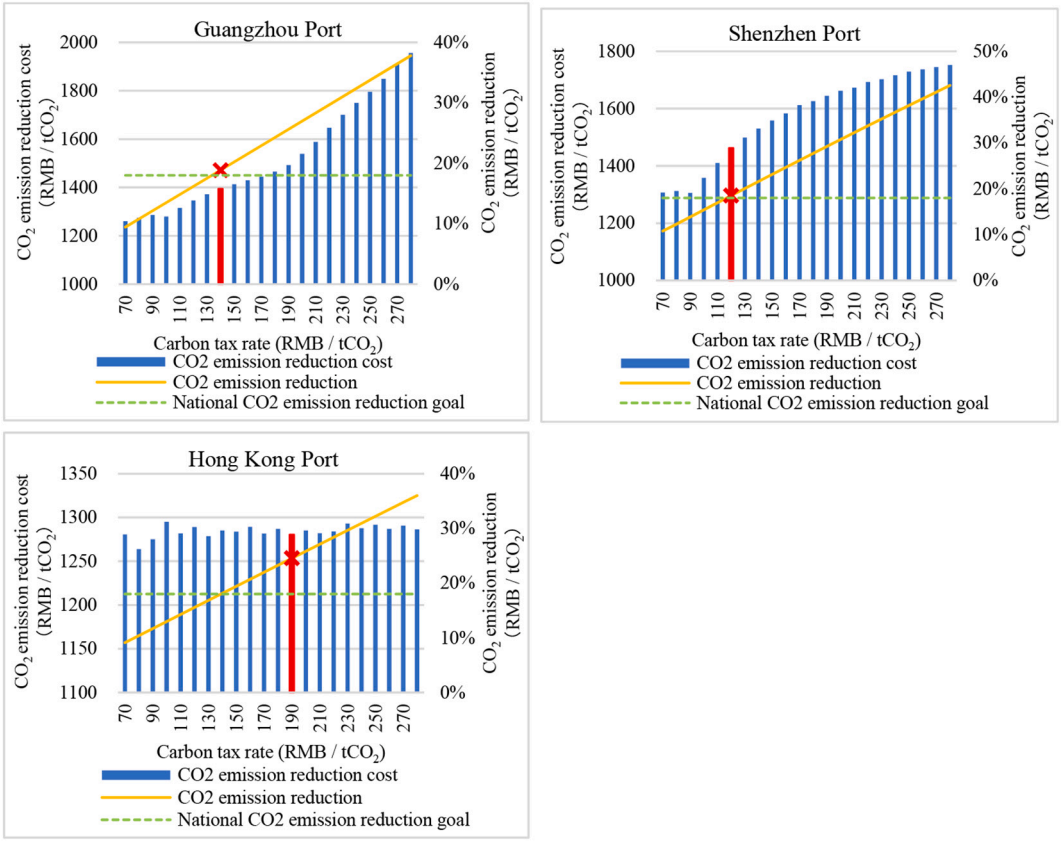


Fig. 5. CO_2 emission reduction costs and emission reduction effects under different carbon tax rates. (Source: compiled by the author based on the simulation results.)

Table 7
The Carbon Tax Rate of Different Ports to Achieve the National Emission Reduction Goal.

Port	Carbon Tax rate (RMB / tCO_2)	CO_2 Emission Reduction Cost (RMB / tCO_2)	CO_2 Emission Reduction
Guangzhou	140	1394	18.9%
Shenzhen	120	1462	18.5%
Hong Kong	190	1280	24.5%

Source: compiled by the author based on the simulation results.

4.1.3. *Impact of carbon taxation on CO_2 emissions*

In the LCT, MCT and HCT scenario, CO_2 emissions from port hinterland intermodal transport will be reduced by 1–2%, 9–11% and 36–44% during 2021–2025 (Table 6). The simulation result shows that changes in carbon tax at a higher rate has a more significant effect on emissions reduction, compared with a low rate. The low carbon tax is not effective in reducing CO_2 emissions, but the medium carbon tax and high carbon tax will lead to a considerable reduction in CO_2 emission. The reason for this is that the CO_2 emission of road transportation is much higher than that of railway transportation and water transportation. The medium carbon tax and high carbon tax will effectively induce the port hinterland transportation modal to change from road transportation to railway and waterway transportation. Therefore, the medium and high carbon tax policy can effectively reduce the CO_2 emissions of port hinterland intermodal transport by reducing the volume of road transportation and promoting the development of railway and waterway transportation.

4.2. *Carbon tax rate discussion*

According to the "14th Five-Year Plan for National Economic and Social Development of the People's Republic of China and the Outline of Long-Term Goals for 2035", the national CO_2 emission reduction goal is to reduce CO_2 emissions per unit of GDP by 18% during 2021–2025. Based on the simulation results of the port hinterland transport CO_2 emissions in NCT, LCT, MCT and HCT scenario, it can be found that the carbon tax rate needs to be set at the medium to high level (70–280 RMB/tCO_2) to achieve the national emission reduction goal.

In the following discussion, this study evaluates the carbon tax rate of different ports to achieve the national emission reduction goal with the lowest emission reduction cost, as shown in Fig. 5. Under the same carbon tax rate, the emission reduction effect in descending order is Shenzhen Port, Guangzhou Port, and Hong Kong Port, which means Shenzhen Port is the most sensitive to carbon tax rate due to its huge emission reduction potential. It can be found that the carbon tax rate of Guangzhou Port, Shenzhen Port and Hong Kong Port need to be set 140, 120 and 190 RMB/tCO_2 to achieve the national emission reduction goal with the lowest emission reduction cost. Besides, in order to jointly achieve the national goal of reducing CO_2 emissions, Shenzhen must pay more than Guangzhou and Hong Kong (Table 7).

5. Conclusion

In this study, we constructed the port hinterland intermodal transport SD model of the GBA. Through an authenticity test and a dimensional consistency test, we demonstrated that the SD model is reliable and effective and simulated the changes in port container throughput, hinterland intermodal transportation and carbon dioxide emissions in NCT scenario, LCT scenario, MCT scenario and HCT scenario.

It can be found that Shenzhen Port is the most sensitive to carbon tax policies. After the implementation of the carbon tax policy, it can be found that the reduction of Shenzhen Port far exceeds that of Guangzhou Port and Hong Kong Port. In the NCT scenario, it can be found that the port hinterland intermodal transport in the GBA is mainly by road and waterway, and rarely by railway. Moreover, even in the same sea port, the share rate of port hinterland transport modals in different port areas is quite different. Base on the simulation results of LCT, MCT and HCT scenario, it can be found that the carbon taxation policy has a depressing effect on road transportation. Comparing the five port areas in the model, the greater the share of road transportation in the port hinterland, the more obvious the inhibitory effect of the carbon taxation policy. The low carbon tax is not effective in reducing CO_2 emissions, but a medium carbon tax and a high carbon tax will lead to a considerable reduction in CO_2 emission.

In terms of port hinterland intermodal transport structure optimization and CO_2 emission reduction, road transportation will inevitably become the focus of any process aimed at achieving a low-carbon port hinterland intermodal transport system in the future. The GBA should rely primarily on the green development of the port hinterland intermodal transport system with a focus on rail and water transit. To achieve the national emission reduction goal, the carbon tax rate needs to be set at the medium to high level. As environmental protection and sustainable development are the common tasks of the port clusters in the GBA, all ports have the responsibility and obligation to reduce CO_2 emissions from port hinterland intermodal transport. However, due to the differences in the main port hinterland transport modal and CO_2 emissions of the ports, each port should set differentiated carbon tax rate. From the perspective of the overall development of the GBA Area, the optimal carbon tax rate should be set at 120–130 RMB/tCO_2. It can be found that the cost of emission reduction by Shenzhen Port is much higher than that of Guangzhou Port and Hong Kong Port. For Shenzhen Port, whose main hinterland transport modal is road, it is the most sensitive to carbon tax rate due to its huge emission reduction potential. While the main hinterland transport modal of Guangzhou Port and Hong Kong Port is waterway. It can be found that the carbon tax rate of Guangzhou Port, Shenzhen Port and Hong Kong Port need to be set 140, 120 and 190 RMB/tCO_2 to achieve the national emission reduction goal with the lowest emission reduction cost. Besides, in order to jointly achieve the national goal of reducing CO_2 emissions, Shenzhen must pay more than Guangzhou and Hong Kong. In promoting the GBA port cluster to meet the emission reduction requirements of the 14th Five-Year Plan period of China, all ports should share responsibility, but not set unified emission reduction targets. Shenzhen Port needs to set higher emission reduction targets (19%–20%), while Guangzhou Port and Hong Kong Port can have emission reduction targets slightly lower than the national emission reduction targets (16%–18%).

The carbon taxation policy is a relatively direct and efficient incentive for inducing the port hinterland transport modal to change from road transportation to railway and waterway transportation, which effectively reduces the CO_2 emissions from port hinterland intermodal transport. To maximize the effect of the carbon taxation policy, a reasonable carbon tax rate should be set while taking into account environmental and economic benefits. If only relying on the carbon taxation policy to achieve the national emission reduction target, the government has to set carbon tax rate at the medium-high level, which will restrict the economic development. The government may consider combining a variety of policy means to promote and optimize the structure of Port hinterland intermodal transport, such as accelerating the construction and development of inland railway and inland waterway infrastructure, rectifying of illegal overloading phenomenon in road freight transportation, providing financial subsidies for inland waterway and railway transport, promoting and regulating the carbon emission trading markets, and so on.

The statement of contribution

Huiling ZHONG: Conceptualization, Supervision, Funding Acquisition;

Wenhua CHEN: Methodology, Writing & Revising - Original Draft;

Yimiao GU: Supervision, Review, Writing Revised Version.

Declaration of Competing Interest

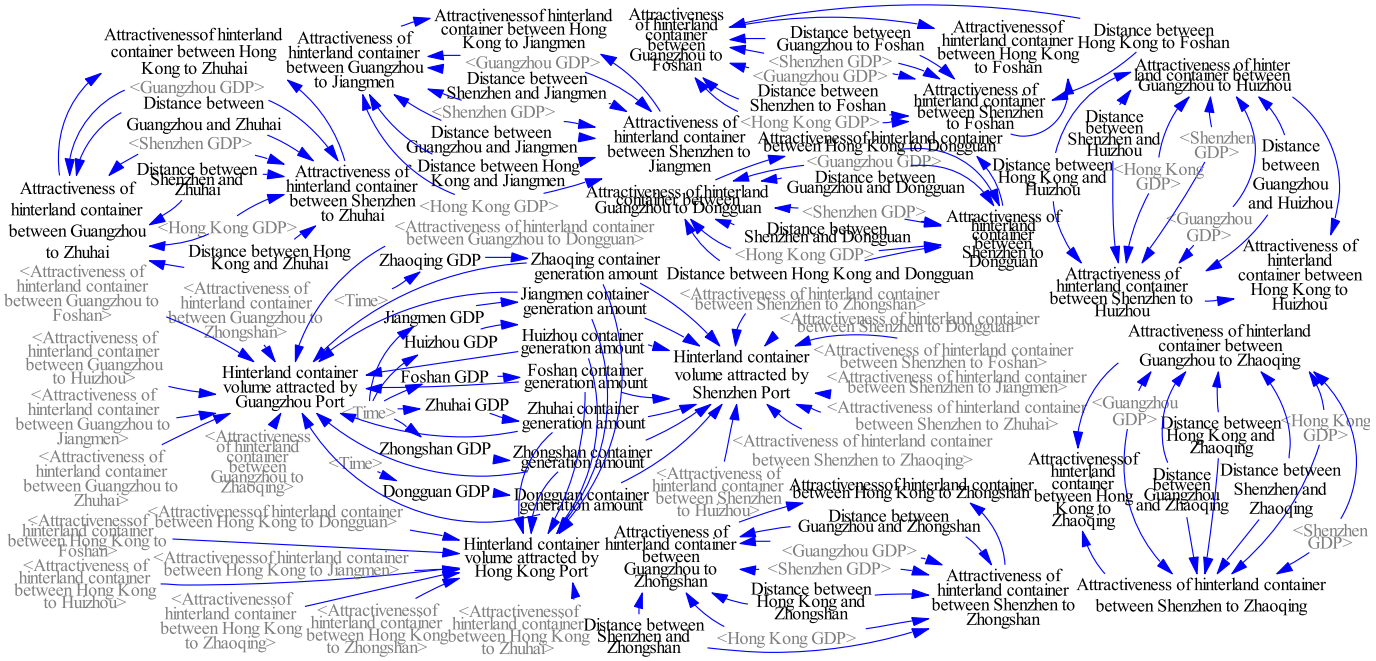
None.

Acknowledgments

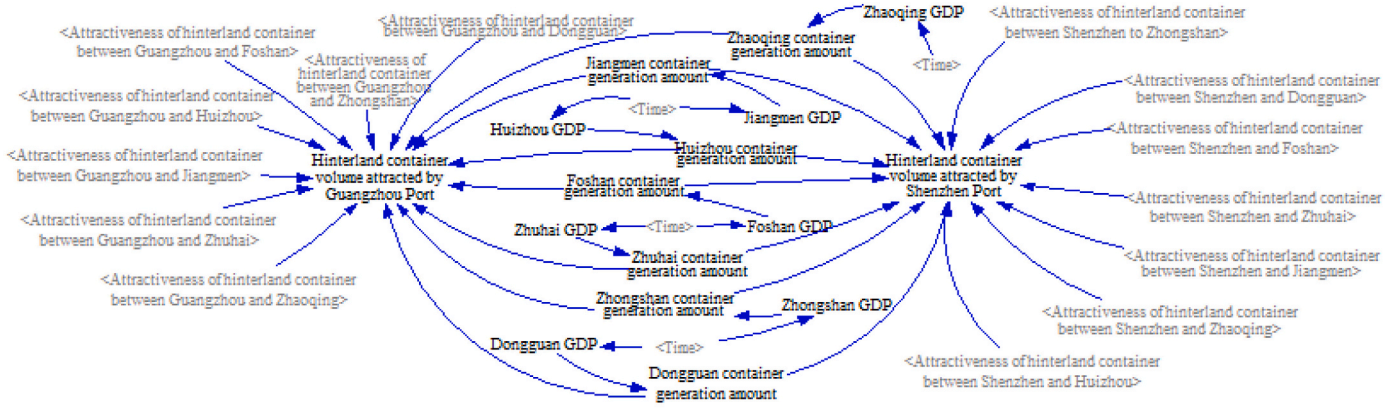
This research is supported by the National Social Sciences Foundation of China (20BGL258).

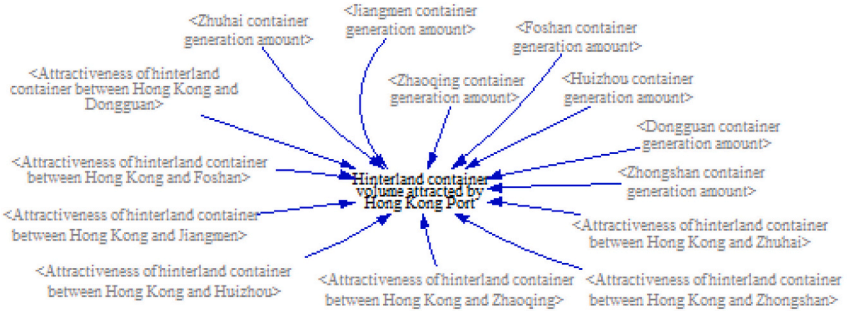
We acknowledge the two anonymous referees for their valuable comments and suggestions.

Appendix A. The flow diagram of hinterland economy subsystem (Integrated diagram)

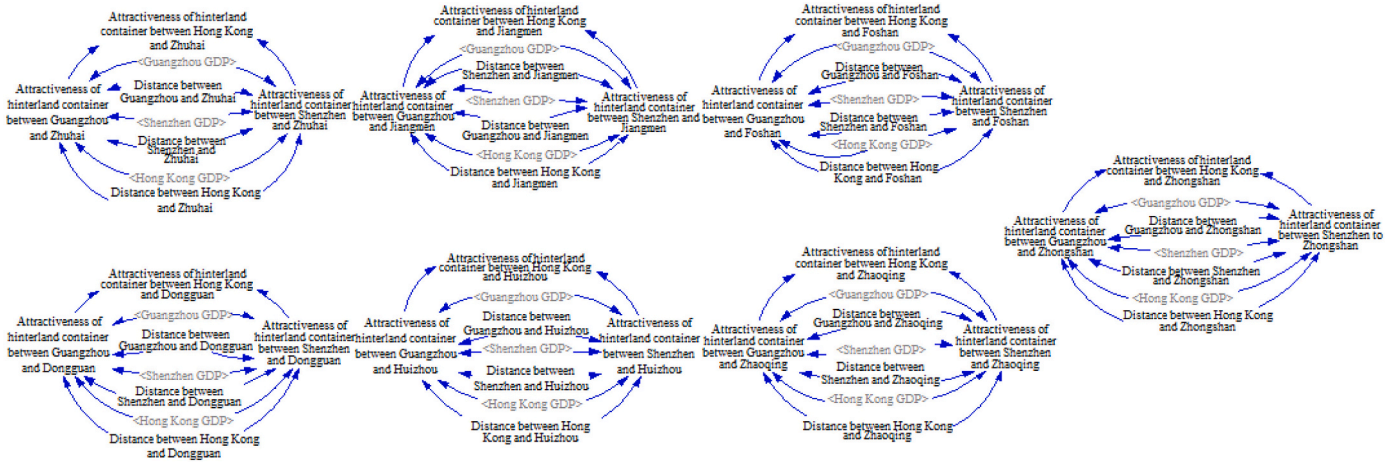


Appendix B. The flow diagram of hinterland economy subsystem (from three ports' perspectives)





Appendix C. The flow diagram of hinterland economy subsystem (from seven cities' perspectives)



Appendix D. The flow diagram of port hinterland intermodal transport subsystem (from different port areas' perspectives)

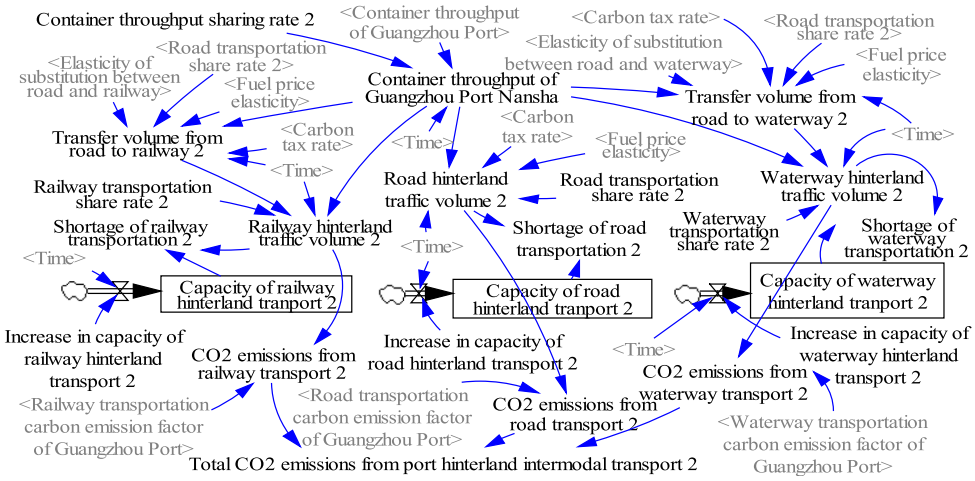


Fig. D.1. The flow diagram of Guangzhou Port (Nansha) hinterland intermodal transport subsystem.

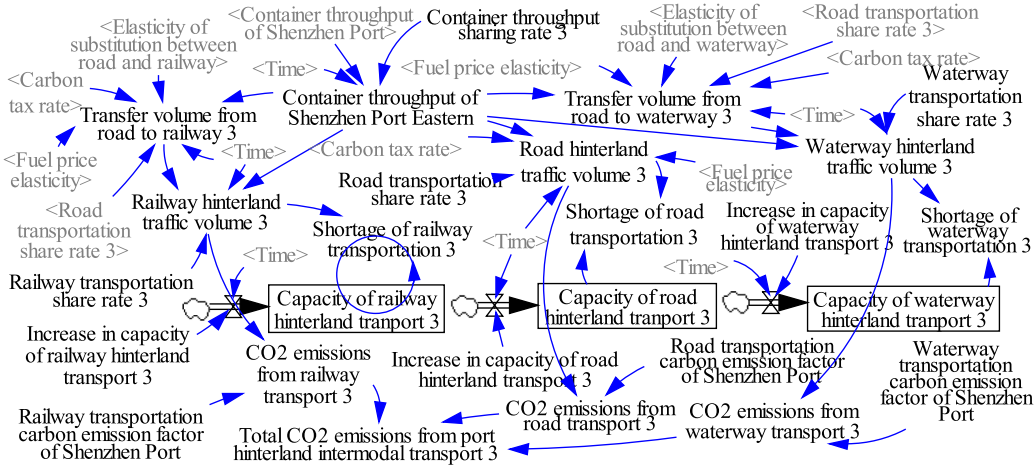


Fig. D.2. The flow diagram of Shenzhen Port (Eastern) hinterland intermodal transport subsystem.

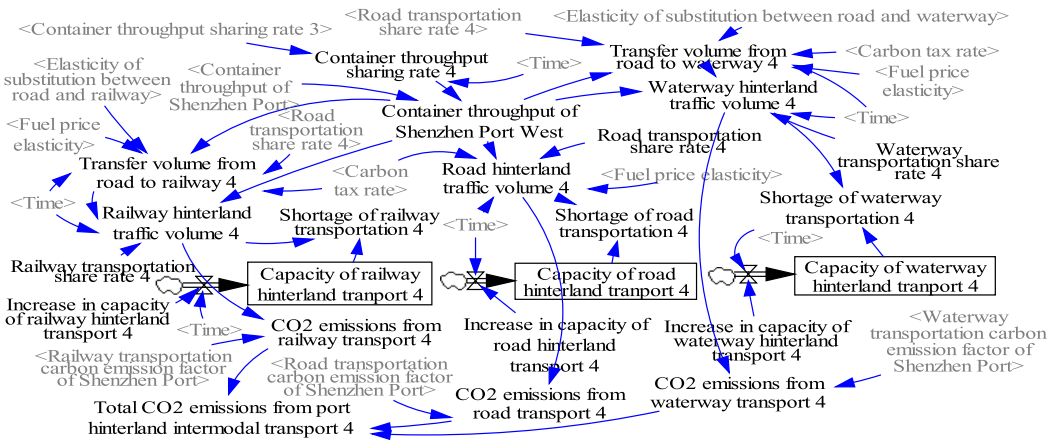


Fig. D.3. The flow diagram of Shenzhen Port (Western) hinterland intermodal transport subsystem.

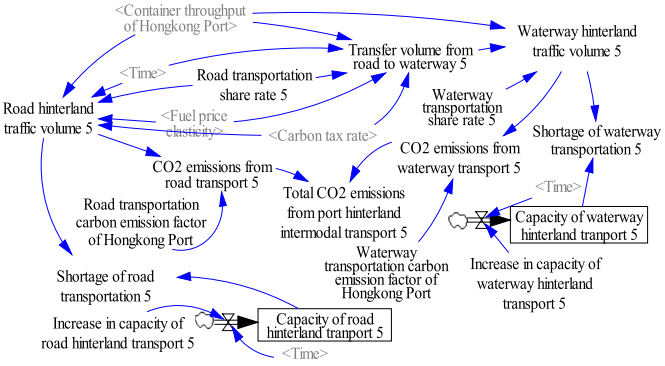


Fig. D.4. The flow diagram of Hongkong Port hinterland intermodal transport subsystem.

Appendix E. The flow diagram of urban economy subsystem (from three port cities' GDP perspectives)

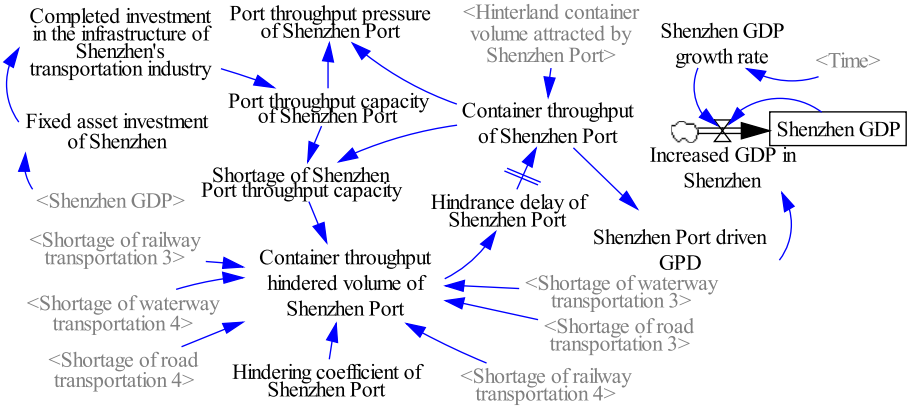


Fig. E.1. The flow diagram of Shenzhen urban economy subsystem.

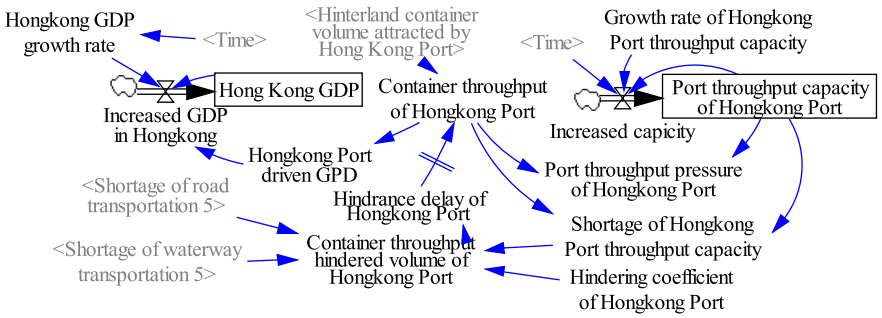


Fig. E.2. The flow diagram of Hongkong urban economy subsystem.

Appendix F. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rtbm.2023.100987.

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